Name: ______________________ **Date:** ______________________

Collaborators: ______________________ **Lab Section:** ______________________

Postlab Exercises (20 points)

Submit the postlab to the TA at the end of the lab.

Moment of Inertia and the Parallel Axis Theorem (10 points)

Question 3 **3 points**

Fill in data from your measurements of the moment of inertia in Data Table 1 and test of the parallel axis theorem in Data Table 2.

Data table 1: *Data for the moments of inertia.*

Ring Measurements

Ring mass [kg] = ______________________

Inner radius R_1 [m] $=$ ______________________

Outer radius R_2 [m] $=$ ______________________

Disk Measurements

Disk mass [kg] = ______________________

Disk radius R [m] $=$ ______________________

Trial	T_{tab} [s]	$T_{\text{tab+ring}}$ [s]
1		
2		
3		
4		
5		
Average [s]		

Data table 2: *Data for parallel axis theorem.*

Distance [m]	$d_{x_1} = 0.000$	$d_{x_2} =$	$d_{x_3} =$	$d_{x_4} =$	$d_{x_5} =$
Trial	T_{x_1} [s]	T_{x_2} [s]	T_{x_3} [s]	T_{x_4} [s]	T_{x_5} [s]
1					
2					
3					
4					
5					
Average [s]					

Next, calculate all values in Data Table 3. For the measured periods, use the average values from Data Tables 1 and 2. Write the formula for each quantity (except for I_{x_2} to I_{x_5}) and the uncertainty. Remember to include units. You do not have to show your calculations.

Data table 3: *Calculation of moments of inertia.*

Formula:	Theoretical value of the moment of inertia of the brass ring (Table 1). Note: we consider it a thick (and heavy) ring.
$I_{\text{ring}} =$	
Formula:	Spring constant of the table, eq. (11), where $I_{\text{obj}} = I_{\text{ring}}$ and $T_{\text{tab+obj}} = T_{\text{tab+ring}}$.
$\kappa =$	
Formula:	Calculated moment of inertia of the table, eq. (13), where $I_{\text{obj}} = I_{\text{ring}}$.
$I_{\text{table}} =$	
Formula:	Theoretical value of the moment of inertia of the brass disk (Table 1).
$I_{\text{disk}} =$	
Formula:	Calculated moment of inertia when the brass disk is placed at d_1. Use eq. (14), where $T_{\text{tab}+x} = T_{x_1}$.
$I_{x_1} =$	
$I_{x_2} =$	Calculated moment of inertia when the brass disk is at d_2.
$I_{x_3} =$	Calculated moment of inertia when the brass disk is at d_3.
$I_{x_4} =$	Calculated moment of inertia when the brass disk is at d_4.
$I_{x_5} =$	Calculated moment of inertia when the brass disk is at d_5.
Uncertainty $= \lvert I_{\text{disk}} - I_{x_1} \rvert =$	Calculated uncertainty in the moment of inertia of the solid disk.

Question 4 **2 points**

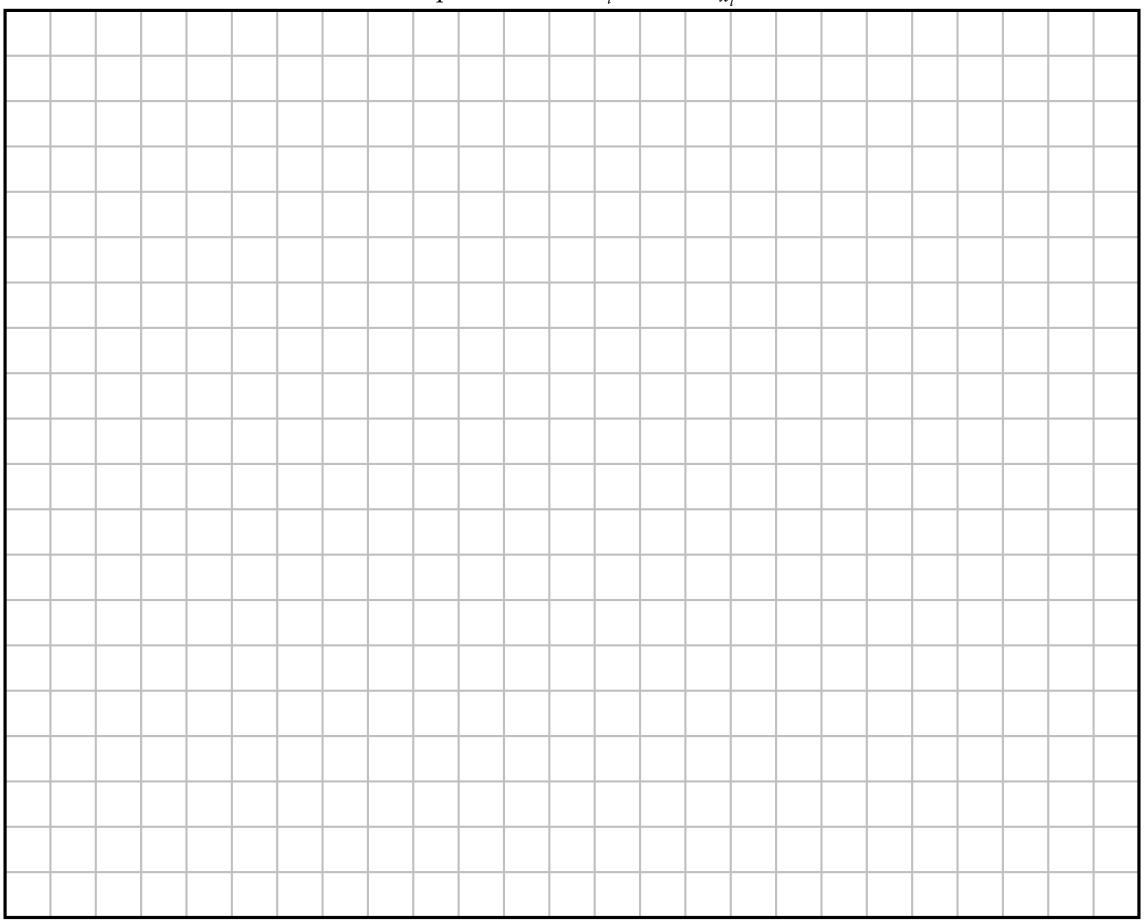
In Graph 1:

(a) (1 point) Plot I_{x_i}, the moment of inertia of the solid disk on the y-axis versus $d_{x_i}^2$, the *square* of the distance between the center of the table and the center of the solid disk, on the x-axis.

(b) ($^1\!/_2$ point) Include error bars on each data point. Each error bar should have the same length $|I_{\text{disk}} - I_{x_1}|$ from Data Table 3.

(c) ($^1\!/_2$ point) Add a title to the plot, label the axes, and include units.

Graph 1: *Plot of* I_{x_i} *versus* $d_{x_i}^2$.



Question 5 **1 points**

(a) (1/2 point) In the plot, draw a best-fit straight line for your data. It need not go through the origin.

(b) (1/2 point) Find the y-intercept of your best-fit line and circle it on the graph. Then circle two points on your best fit line, (x_1, y_1) and (x_2, y_2), and compute the slope. Write both values below:

$$b_{\text{exp}} = y - \text{intercept} =$$

$$m_{\text{exp}} = \text{slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} =$$

Question 6 **1 point**

If you take the parallel axis theorem

$$I = Md^2 + I_{\text{CM}}$$

and plot I versus d^2 (y vs. x), the data should be linear. Given that the equation for a line is $y = mx + b$, where m is the slope and b is the y-intercept, what variables in the parallel axis theorem correspond to m and b?

$$m_{\text{theory}} = \text{slope} =$$

$$b_{\text{theory}} = \text{intercept} =$$

Question 7 **2 points**

Using your answers to questions 5 and 6, how much do your measured values for the slope and y-intercept differ from the theoretical values? Remember to include units.

$$|m_{\text{theory}} - m_{\text{exp}}| =$$

$$|b_{\text{theory}} - b_{\text{exp}}| =$$

Question 8 **1 points**

Name two sources of experimental error. Give a reason as to how each source of error would affect your values of the moment of inertia.

(a) ($^1/_2$ point) First source of error:

(b) ($^1/_2$ point) Second source of error:

Hooke's Law and Spiral Spring Oscillations (10 points)

Add your measurements into Data Table 4 below:

Data table 4: *Hooke's Law and Spiral Spring measurements.*

m = mass of spring [kg] =

M = attached mass [kg]	Force = Mg [N]	Extension [m]	Period [s]	Period^2 [s^2]
0.005				
0.010				
0.015				
0.020				
0.025				

Question 9 **2 points**

(a) (1 point) Plot the **force** of gravity versus **extension** (y vs. x) in Graph 2 using the data from Data Table 4.

(b) ($^1/_2$ point) Include a plot title, axis labels, and units.

(c) ($^1/_2$ point) Draw a best-fit straight line through the data points.

Graph 2: *Force of gravity versus spring extension.*

Question 10 **1 point**

Based on the best-fit straight line, does your data agree with Hooke's Law? Explain why.

Question 11 **1 point**

As in question 5, calculate the slope of the best-fit straight line. Circle two points (x_1, y_1) and (x_2, y_2) on the line that you will use to compute the slope, and use it to find the spring constant, k_{Hooke}. Don't forget units!

$$k_{\text{Hooke}} = \text{slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} =$$

Question 12 **1 point**

If we were to plot T^2 along the y-axis and M along the x-axis:

(a) (1/2 point) What would be an equation for the theoretical value of the slope, m_{theory}, remembering that $y = mx + b$ is the equation for a line?

(b) (1/2 point) What would be an equation for the theoretical value of the y-intercept, b_{theory}?

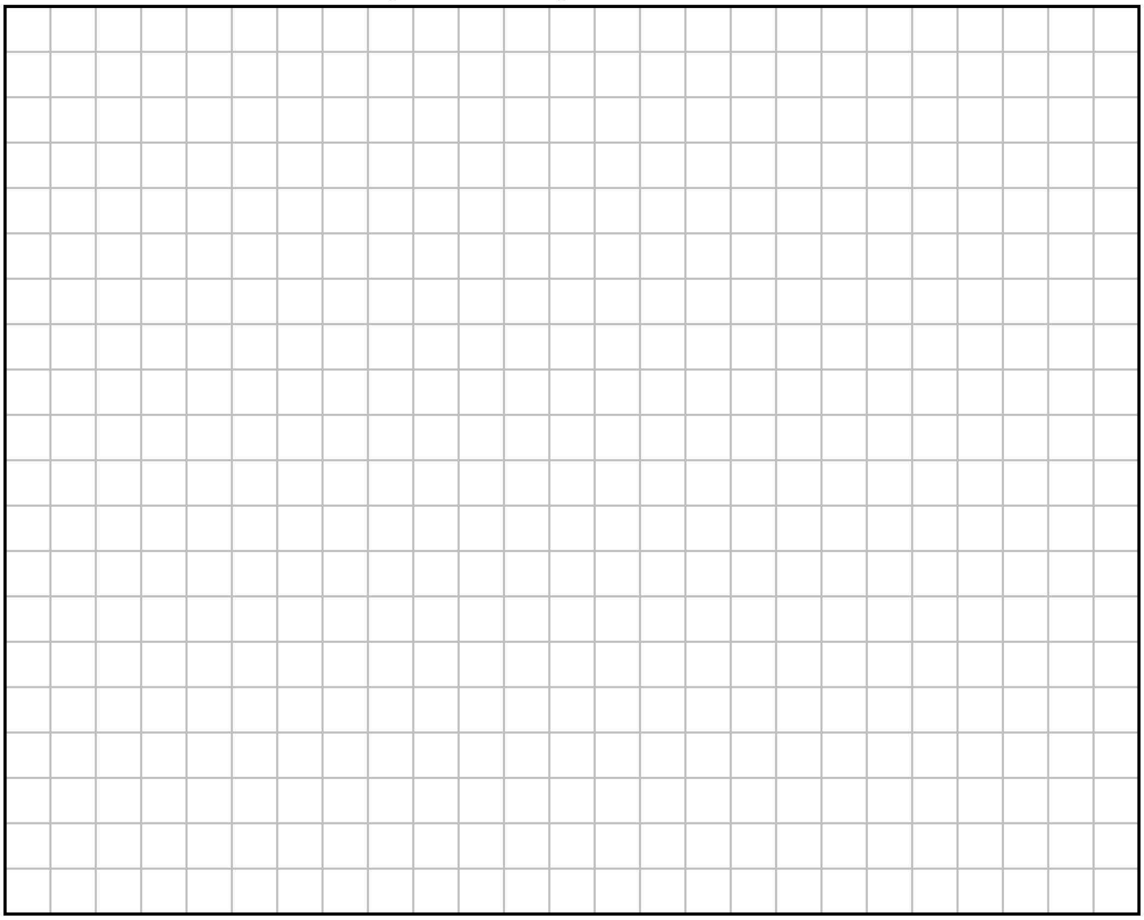
Question 13 **1 point**

Using your data for Period^2 and the attached mass M from Data Table 4:

(a) ($^1/_2$ point) Plot Period^2 versus attached mass, i.e., T^2 vs M, in Graph 3.

(b) ($^1/_2$ point) Draw a best-fit straight line through the data, and include a title, axis labels, and units.

Graph 3: *Period squared versus mass.*



Question 14 **1 point**

Calculate the slope of the best-fit straight line. Circle two points on the line, (x_1, y_1) and (x_2, y_2). Setting the measured slope equal to the theoretical expression you found in question 12, find the spring constant k_{harm}:

$$m_{\text{exp}} = \text{slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = m_{\text{theory}}$$

$$k_{\text{harm}} =$$

Question 15 **1 point**

Find and circle the y-intercept of the best-fit straight line. Call this intercept y_0. Setting y_0 equal to the theoretical expression for the y-intercept from question 12, find the spring mass coefficient b. Use the spring constant k_{harm} you computed in question 14.

$y_0 = \text{intercept} =$

$b =$

Question 16 **1 point**

Suppose that if the effective spring mass, bm, is less than 10% of the smallest attached mass, $M = 0.005\,\text{kg}$, then we can say the spring mass is negligible. Based on the value of b found in question 15, is the spring mass negligible? That is, do you find that

$$\frac{bm}{M} \leq 0.1?$$

Question 17 **1 point**

Check whether the spring constants k calculated in questions 11 and 14 are comparable in two steps:

(a) ($^1\!/_2$ point) Find the difference between the two k values, including units:

$|k_{\text{Hooke}} - k_{\text{harm}}| =$

(b) ($^1\!/_2$ point) Suppose we assume the k measurements are consistent if they differ by less than 10% of the smaller of the two values. Is it the case that

$$\frac{|k_{\text{Hooke}} - k_{\text{harm}}|}{k_{\text{smaller}}} \leq 0.1?$$